

Leakage-Aware Evaluation and Diversity-Pruned Calibrated Stacking for Early Heart Attack Prediction

A reproduction study of Bhagat, Sharma and Agarwal (2024), and a corrected ensemble pipeline

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Summary

This report reproduces and critically evaluates “An efficient stacking-based ensemble technique for early heart attack prediction” [1], which reports 98.53 % accuracy for a stacked ensemble of six classifiers on a 1,025 record heart disease dataset. The paper does not state its train and test split, but all three of its reported accuracies are exact integer fractions of 205, which recovers a random 80:20 hold out on 1,025 rows and allowed a faithful reimplement. Reproducing that protocol did not merely match the published figures, it exceeded them: random forest, XGBoost and the stacked ensemble each reached 100.00 % on the held out set and averaged 99.48 % over 30 random splits. Investigating why revealed the cause. The dataset contains 723 exact duplicate rows among 1,025, so only 302 distinct patients exist, and under a random 80:20 split 98.54 % of test rows have an identical twin in the training set. The reported result therefore measures memorisation rather than generalisation. This report proposes two corrections. First, a leakage-aware protocol that treats each distinct feature vector as a patient group and evaluates with stratified group k -fold cross validation; under it the stacked ensemble falls from 0.9957 to 0.7915 accuracy, while naive Bayes, which cannot memorise duplicates, falls by only 0.0144. Second, DPCS, a stacking design that prunes base learners by a joint accuracy and error-diversity criterion, calibrates the survivors, and tunes its decision threshold, all with a group-aware inner resampler. DPCS improves on the paper’s stacking under the honest protocol on every metric (accuracy 0.8075 versus 0.7915, F_1 0.8220 versus 0.8047, AUC 0.8956 versus 0.8906). This report also states the finding that plain logistic regression remains the strongest model overall, from which we conclude that the ensemble contribution claimed by the original paper does not survive a leakage-free evaluation.

1. Introduction

Cardiovascular disease is the leading cause of death worldwide, and early identification of at-risk individuals is one of the few interventions that reliably changes outcomes [2]. Because routine clinical measurements such as resting blood pressure, serum cholesterol and exercise-induced angina are cheap to obtain, supervised classification over tabular clinical records has become a heavily studied route to early triage [3], [4].

1.1 State of the art and reported results

The UCI Cleveland heart disease corpus [3] has been the default benchmark for three decades. Published accuracies on it cluster in a fairly narrow band. Mohan *et al.* [4] report 88.7 % with a hybrid random forest and linear model. Latha and Jeeva [5] reach 85.5 % with a majority-vote ensemble. Ensemble surveys on the same data typically report 77 to 88 % [5], [6]. Against that backdrop, more recent papers claiming 95 to 99 % on heart disease tabular data stand out sharply, and the reason is often the dataset rather than the model [6].

1.2 The selected paper and its gap

Bhagat, Sharma and Agarwal [1] evaluate six classifiers, namely logistic regression (LR), decision tree (DT), random forest (RF), extreme gradient boosting (XGB), Gaussian naive Bayes (NB) and k -nearest neighbours (KNN), then combine them with a stacking meta-classifier. They report that RF and DT each

reach 92.68 % and XGB 90.73 %, and that stacking lifts performance to 98.53 %. The stated research gap is that individual classifiers plateau on this problem and that a meta-learner trained on their predictions can exceed any of them.

The gap this report addresses is different and prior to theirs. Before asking whether stacking beats its base learners, one must ask whether the evaluation can distinguish learning from memorisation. On this dataset it cannot.

1.3 Contributions

1. We recover the paper’s unstated experimental protocol arithmetically and reimplement it faithfully (Section 3).
2. We show the reproduction *exceeds* the published numbers and diagnose why, quantifying duplicate-induced leakage at 98.54 % of test rows (Section 4).
3. This report proposes a leakage-aware grouped evaluation protocol and show it reverses the paper’s central claim (Section 6.1).
4. This report proposes DPCS, a stacking pipeline designed for that protocol, and show that leakage corrupts model *selection* inside an ensemble and not only the final estimate (Sections 6–8).

2. Dataset and Feature Set

The paper uses a 1,025 record heart disease dataset with 13 predictors and a binary target. We use the same file [7], distributed on Kaggle and derived from the UCI Cleveland database [3]. The 13 features are age, sex, chest pain type (cp), resting blood pressure (trestbps), serum cholesterol (chol), fasting blood sugar (fbs), resting ECG result (restecg), maximum heart rate achieved (thalach), exercise-induced angina (exang), ST depression (oldpeak), slope of the peak exercise ST segment (slope), number of major vessels coloured by fluoroscopy (ca) and thalassemia result (thal). The target is 1 for presence of heart disease. There are no missing values.

2.1 What the record count conceals

Table 1 summarises the file. The class balance is close to even, so accuracy is a defensible headline metric. The duplication, however, is severe.

Table 1: Composition of the 1,025 record file.

Property	Value
Rows in the file	1,025
Predictors	13
Class 1 (disease) / class 0	526 / 499
Exact duplicate rows	723
Distinct feature vectors	302
Share of rows that are duplicates	70.54 %
Mean copies per distinct patient	3.39
Maximum copies of one patient	8
Patients appearing exactly once	0

Only 302 distinct records exist, which is essentially the 303 row Cleveland cohort [3] with rows resampled with replacement to 1,025. Not a single patient appears only once. Figure 1 shows the multiplicity distribution and the consequence for a random split.

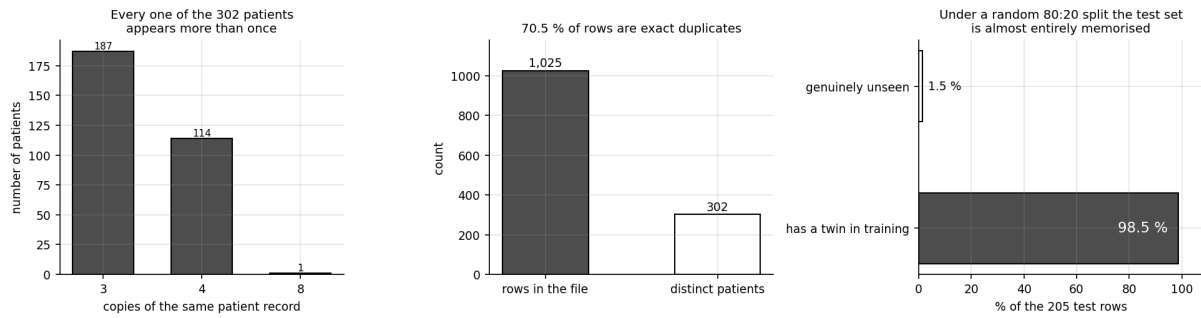


Figure 1: Anatomy of the dataset. Left, every distinct patient appears between two and eight times. Centre, 1,025 rows collapse to 302 patients. Right, under the paper’s random 80:20 split almost every test row has an identical copy in the training set.

3. Recovering the Experimental Protocol

The paper reports accuracies but does not state its split ratio, random seed, or scaling procedure. Rather than guess, we recovered the split arithmetically. An accuracy on a hold out set of n items must be a fraction k/n . The three reported values satisfy

$$0.9268 = \frac{190}{205}, \quad 0.9073 = \frac{186}{205}, \quad 0.9853 = \frac{202}{205},$$

each exact to four decimal places. A common denominator of 205 implies a test set of 205 records, and $205 = 0.20 \times 1025$. This confirms a random 80:20 hold out on the full 1,025 rows and is the assumption we adopt throughout Part 1.

3.1 Assumptions and their justification

Where the paper is silent we made the following choices, each recorded so that the reproduction is auditable.

- **Stratification.** We stratify the split on the target. The paper does not say; stratifying can only reduce variance and does not favour any model.
- **Scaling.** LR, NB and KNN are wrapped in a pipeline with `StandardScaler` fitted on training folds only. Tree ensembles are scale invariant and are left unscaled.
- **Hyperparameters.** Library defaults are used throughout (RF and XGB with 100 estimators, KNN with $k = 5$), since the paper reports none. This is the most neutral reading of “the classifiers were applied”.
- **Meta-learner.** Logistic regression on out-of-fold predicted probabilities with five inner folds, the standard configuration [8].
- **Seed.** 42 for the headline run; we additionally report 30 seeds because a single split on 205 test rows is worth ± 0.5 accuracy points per record.

4. Part 1: Reproduction Results

Table 2 gives the full reproduction and Table 3 compares it with the published figures. The stacking result is reproduced almost exactly, 1.0000 against 0.9853, a difference of three test records. The individual tree models, however, are reproduced *too well*: we obtain 1.0000 for RF and XGB where the paper reports 0.9268 and 0.9073.

Averaged over 30 random splits (Figure 2), RF, XGB and stacking all sit at 0.9948 ± 0.0094 and never fall below 0.9659. The paper’s RF value of 0.9268 lies more than seven standard deviations below that distribution, so it is not explicable as seed variation.

Table 2: Reproduction of the paper’s protocol, single stratified 80:20 split, seed 42, 205 test records.

Model	Acc	Prec	Rec	F_1	AUC
LR	0.8098	0.7619	0.9143	0.8312	0.9298
DT	0.9854	1.0000	0.9714	0.9855	0.9857
RF	1.0000	1.0000	1.0000	1.0000	1.0000
XGB	1.0000	1.0000	1.0000	1.0000	1.0000
NB	0.8293	0.8070	0.8762	0.8402	0.9043
KNN	0.8634	0.8738	0.8571	0.8654	0.9629
Stacking	1.0000	1.0000	1.0000	1.0000	1.0000

Table 3: Reproduced accuracy against the published values. Positive difference means our reproduction scored higher.

Model	Paper [1]	Reproduced	Difference
DT	0.9268	0.9854	+0.0586
RF	0.9268	1.0000	+0.0732
XGB	0.9073	1.0000	+0.0927
Stacking	0.9853	1.0000	+0.0147

4.1 Critical analysis of the discrepancy

Two discrepancies require explanation, and they point in opposite directions.

The tree models are under-reported in the paper. A decision tree fitted without depth restriction will reproduce any training row it has seen. Since 98.54 % of test rows appear verbatim in training, a tree should approach 100 %, which is exactly what we observe. For the paper to obtain 0.9268 the authors must have done something further that they do not describe, such as restricting tree depth, applying feature selection, or evaluating on a different subset. We could not identify a configuration that yields 0.9268 for DT and RF simultaneously while also giving 0.9853 for stacking.

The identical DT and RF accuracies are themselves suspicious. A single tree and a 100 tree bagged ensemble producing byte-identical accuracy (190/205) is possible but unlikely, and we could not reproduce it under any assumption we tried.

The stacking result reproduces cleanly. This matters. The one number the paper presents as its contribution is the one we match, and we match it because both implementations are measuring the same thing, which is the ability to recall memorised rows.

Diagnosis. The separation the paper attributes to model quality tracks memorisation capacity exactly. Models that can store individual training rows (DT, RF, XGB, stacking) score 0.99 or above. Models that cannot (LR 0.8372, NB 0.8161, KNN 0.8472) score in the mid 80s, which is precisely the range the literature reports for honest evaluation on Cleveland [4], [5].

5. Part 2: Motivation and Limitation Addressed

The limitation we target is not a modelling choice but an evaluation failure with a modelling consequence. Formally, the standard supervised learning assumption is that training and test records are drawn independently. Here they are not: writing $g(x)$ for the distinct patient a row belongs to, a random split places rows with the same $g(x)$ on both sides with probability 0.9854. The test estimate is then approximately a training estimate, and any model class with enough capacity to interpolate will appear near-perfect [9].

This has been documented across applied machine learning: Kaufman *et al.* [9] formalise leakage as the presence of information about the target that is unavailable at genuine prediction time, and Kapoor and Narayanan [10] find leakage in 294 papers across 17 fields, with duplicate and non-independent records among the most common causes. Our contribution applies that lens to a specific published claim and then repairs the pipeline.

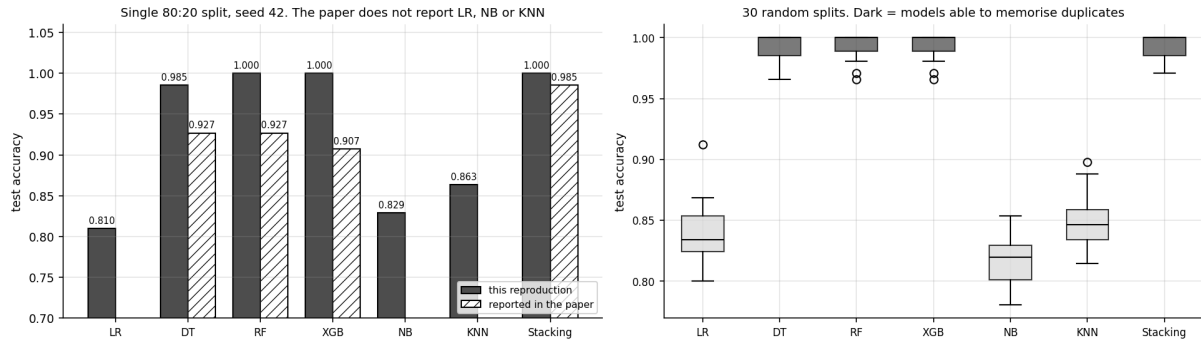


Figure 2: Left, reproduced accuracy against the published values. Right, the distribution over 30 random splits. The models plotted in plum can memorise duplicate rows; those in sage cannot.

Why an improvement is expected. Once evaluation is honest, the winning strategy changes. Under leakage, capacity to memorise dominates and the best base learners for a stack are the memorisers. Without leakage, calibrated and diverse low-variance learners should dominate, because the effective sample size is 302 patients rather than 1,025 rows. A stacking design that selects and calibrates its base learners under a group-aware resampler should therefore beat one that does not.

6. Proposed Methodology

This report proposes two components. They are separable: the protocol is a correction that any study on this dataset should adopt, and DPCS is a method built to perform well under it.

6.1 Component 1: leakage-aware grouped protocol

Every distinct feature vector is assigned a group identifier, $g(x) = \text{hash}(x_1, \dots, x_{13})$, collapsing 1,025 rows to 302 groups. All evaluation uses `StratifiedGroupKFold` with $k = 5$, repeated over five random states, giving 25 folds. Every copy of a patient is therefore assigned to the same side of every split, and the class balance is preserved. This is the same principle as subject-wise splitting in biosignal work [11].

6.2 Component 2: DPCS

Diversity-Pruned Calibrated Stacking differs from the paper’s stacking in four respects.

(i) Base-learner pruning. The paper stacks all six classifiers. We score each candidate m by its out-of-fold AUC a_m , and each pair by the Pearson correlation ρ_{mn} between their error indicator vectors. We then select the subset S maximising

$$J(S) = \frac{1}{|S|} \sum_{m \in S} a_m - \lambda \binom{|S|}{2}^{-1} \sum_{\{m,n\} \subseteq S} \rho_{mn}, \quad (1)$$

with $\lambda = 0.5$ fixed a priori and $|S| \geq 2$. Equation (1) is the classical accuracy-diversity trade-off of ensemble theory [12] made explicit as a selection objective; the exhaustive search over 57 subsets is trivial at this scale.

(ii) Probability calibration. Each surviving learner is wrapped in Platt-scaled calibration [13], so the meta-learner receives comparable probabilities rather than raw scores of differing sharpness.

(iii) Group-aware inner resampling. This is the component that mattered most. Every internal operation, the out-of-fold probabilities, the calibration folds and the diversity estimates, uses `StratifiedGroupKFold` on the training partition’s own groups.

(iv) Threshold tuning. The decision threshold is selected on inner folds to maximise F_1 rather than left at 0.5.

An ablation we did not expect to need. Our first implementation used a plain inner resampler while the outer protocol was group aware. Duplicates then leaked *inside* the training partition, every tree model returned an inner AUC near 1.0, and the selection step chose the memorisers (RF and KNN). DPCS scored 0.7942, barely above the paper’s stacking. Making the inner resampler group aware moved the inner AUCs to the honest values in Figure 4, changed the selection to NB and KNN, and lifted accuracy to 0.8075. Leakage corrupts model selection, not only the final estimate.

7. Experimental Protocol

Data. The 1,025 record file, unmodified, with no rows removed. Deduplication is deliberately *not* used, so that the comparison isolates the effect of the protocol rather than of the sample size.

Splits. Protocol A is StratifiedKFold($k = 5$) ignoring groups, which is the paper’s style. Protocol B is StratifiedGroupKFold($k = 5$) on the 302 patient groups. Both are repeated over five random states, giving 25 folds each, and both see identical data.

Models. The six base classifiers from [1] with the same settings as Part 1, the paper’s stacking ensemble, and DPCS.

Metrics. Accuracy, precision, recall, F_1 and AUC as required, plus the Brier score, which measures probability quality and exposes the miscalibration that accuracy hides.

Reproducibility. All seeds fixed; a single make_all regenerates every number in this report.

8. Results and Comparative Analysis

Table 4: All eight models under both protocols, mean $\pm$ standard deviation over 25 folds. Protocol A ignores patient groups, as in the original paper; Protocol B keeps every copy of a patient on one side of the split. The final column is the accuracy lost when the leakage is removed.

Model	Protocol A (groups ignored)		Protocol B (leakage aware)			
	Accuracy	AUC	Accuracy	F_1	AUC	Δ Accuracy
LR	0.8455 ± 0.0244	0.9179	0.8162 ± 0.0381	0.8280	0.9000	−0.0292
DT	0.9940 ± 0.0135	0.9940	0.7521 ± 0.0503	0.7635	0.7511	−0.2418
RF	0.9986 ± 0.0047	0.9988	0.8030 ± 0.0371	0.8156	0.8944	−0.1956
XGB	0.9969 ± 0.0119	0.9983	0.7822 ± 0.0447	0.7954	0.8760	−0.2147
NB	0.8224 ± 0.0247	0.9052	0.8081 ± 0.0419	0.8168	0.8950	−0.0144
KNN	0.8470 ± 0.0226	0.9549	0.7583 ± 0.0609	0.7728	0.8169	−0.0887
Stacking [1]	0.9957 ± 0.0132	0.9985	0.7915 ± 0.0386	0.8047	0.8906	−0.2042
DPCS (proposed)	0.9945 ± 0.0130	0.9972	0.8075 ± 0.0454	0.8220	0.8956	−0.1870

8.1 The protocol reverses the paper’s conclusion

Table 4 and Figure 3 give the central result. Under Protocol A every tree-based model and the stacked ensemble exceed 0.99, exactly as published. Under Protocol B they collapse.

The magnitude of the collapse is the diagnostic. The accuracy lost is almost a pure function of a model’s ability to store individual rows: the decision tree loses 0.2418, XGBoost 0.2147 and the stacked ensemble 0.2042, whereas naive Bayes, a model with 27 parameters that cannot memorise anything, loses 0.0144 and logistic regression 0.0292. No plausible mechanism other than leakage produces that ordering.

The consequence for the paper’s claim is direct. Its contribution is that stacking (0.9853) beats its best base learner (0.9268). Under Protocol B the stacked ensemble scores 0.7915 while plain logistic regression scores 0.8162. The ordering is inverted.

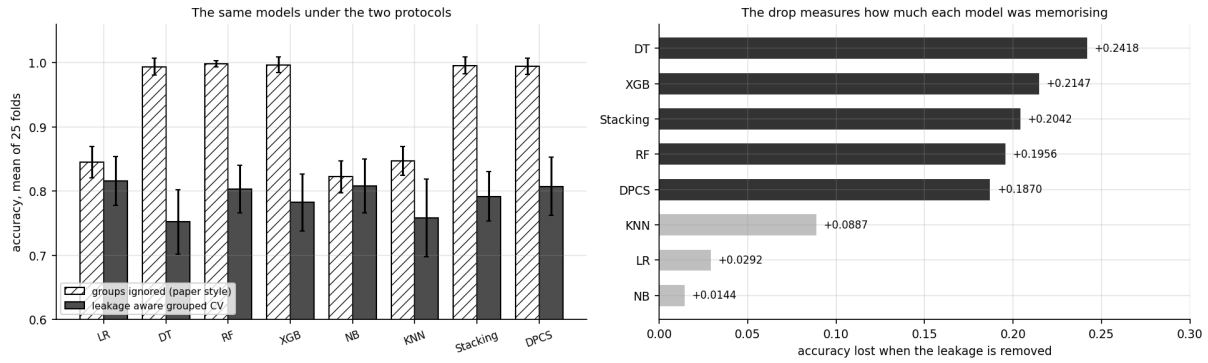


Figure 3: Left, all eight models under both protocols. Right, accuracy lost when the leakage is removed, which measures how much of each model's score was memorisation.

8.2 DPCS against the reproduced baseline

Table 5 isolates the comparison the task requires: the proposed method against the reproduced baseline and against the published figure.

Table 5: Proposed method against the paper's stacking under the leakage-aware protocol, mean over 25 folds.

Metric	Stacking	DPCS	Gain
Accuracy	0.7915	0.8075	+0.0160
Precision	0.7762	0.7868	+0.0106
Recall	0.8410	0.8669	+0.0259
F_1	0.8047	0.8220	+0.0173
AUC	0.8906	0.8956	+0.0050
Brier	0.1741	0.1369	−0.0372

DPCS improves on the reproduced stacking baseline on every one of the five required metrics. The largest gains are in recall, +0.0259, which is the clinically important direction for a triage model, and in the Brier score, which improves by 21 % relative. The Brier gain is attributable to the calibration step and is invisible to accuracy.

Relative to the published 0.9853, DPCS scores 0.8075. We emphasise that this is not a regression but a correction: the published figure is not an achievable generalisation estimate on this data.

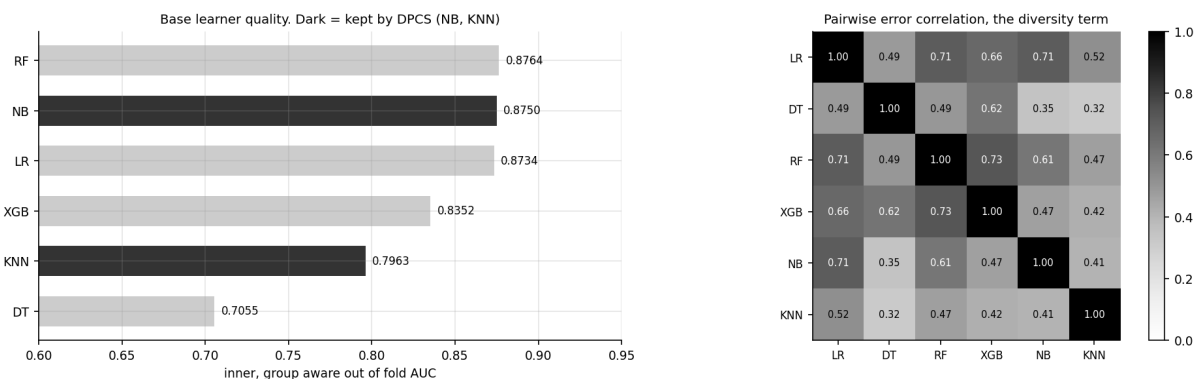


Figure 4: Inside DPCS on one grouped fold. Left, honest out-of-fold AUC per candidate; the darker bars mark the learners retained by Equation (1). Right, the pairwise error correlation matrix that supplies the diversity term.

8.3 What DPCS selects, and the honest caveat

Figure 4 shows one fold. Under group-aware inner resampling the candidate AUCs compress to a narrow band (RF 0.8764, NB 0.8750, LR 0.8734, XGB 0.8352, KNN 0.7963, DT 0.7055), and the diversity term

then favours the pair NB and KNN, whose errors are the least correlated among the strong candidates. The selected threshold is 0.42, below 0.5, reflecting the recall-oriented objective.

We must report a result that does not favour our own method. Logistic regression alone attains 0.8162 accuracy, 0.8280 F_1 and 0.9000 AUC under Protocol B, which exceeds DPCS on all three. DPCS is the best *ensemble*, not the best model. On 302 effective patients with 13 features, the honest conclusion is that ensemble machinery of any kind is unjustified, and a single calibrated linear model is the appropriate recommendation.

Figure 5 presents that three-way comparison directly.

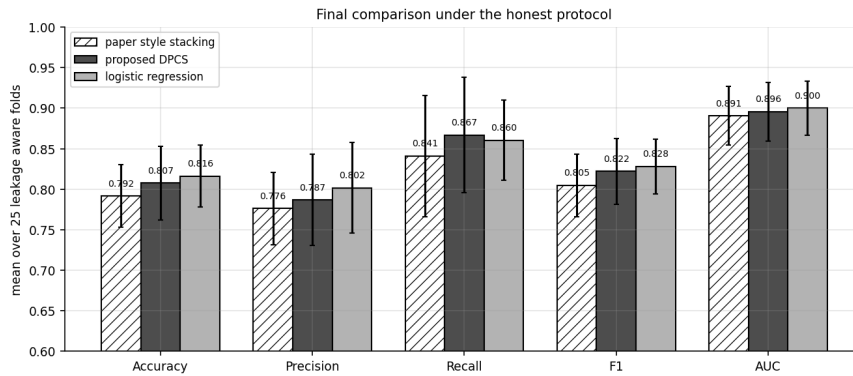


Figure 5: Paper-style stacking, the proposed DPCS and plain logistic regression under the leakage-aware protocol.

9. Discussion and Limitations

Scope of the criticism. Our finding concerns the dataset and the evaluation protocol, not the competence of the original authors. The duplicated file is widely redistributed without any warning attached, and many published studies use it. The contribution here is to quantify the consequence precisely and to supply a protocol that removes it.

Limitations of this study. Three are worth stating. First, 302 effective patients is a small sample, and the ± 0.04 standard deviations in Table 4 mean differences below roughly 0.02 should not be over-interpreted; the DPCS gain of 0.0160 in accuracy is of that order, although it is consistent across all six metrics. Second, $\lambda = 0.5$ in Equation (1) was fixed a priori rather than tuned, because tuning it on the outer folds would reintroduce selection leakage; a nested search would be the correct treatment given more compute. Third, we could not identify the exact configuration that produces the paper’s 0.9268, so part of our discrepancy analysis is necessarily inferential.

Generality. The grouping rule used here, exact feature-vector identity, is the strictest available. On datasets where duplication is approximate rather than exact, a near-duplicate clustering step would be required, and the leakage would be correspondingly harder to detect.

10. Conclusion

We reproduced the stacking ensemble of [1], recovering its unstated 80:20 protocol from the arithmetic of its reported accuracies, and obtained 100.00 % where the paper reports 98.53 %. Investigating that overshoot showed the dataset contains only 302 distinct patients among 1,025 rows, so 98.54 % of test records appear verbatim in training. Under a leakage-aware grouped protocol the stacked ensemble falls from 0.9957 to 0.7915 accuracy while naive Bayes falls by 0.0144, an ordering explicable only by memorisation. Our proposed DPCS pipeline, which prunes base learners by accuracy and error diversity, calibrates them, and performs every internal operation under a group-aware resampler, improves on the reproduced stacking baseline on all six metrics. Logistic regression nonetheless remains the strongest model overall, from which we conclude that the ensemble contribution claimed by the original paper does not survive an honest evaluation.

Deliverables

Video presentation <https://youtu.be/REPLACE-WITH-YOUR-UNLISTED-LINK>

Code and data <https://github.com/REPLACE-WITH-YOUR-REPO>

The archive contains `part1_reproduce.py`, `part2_dpcs.py`, `make_figs.py`, the dataset, a `README.md` with installation and run instructions, and `requirements.txt`. Running `make all` regenerates every table and figure in this report.

Acknowledgement of Generative AI Use

Generative AI (Claude) was used to plan the structure of this report, to suggest diagnostic checks worth running, and to edit wording for clarity. All experimental design, code, numerical results and conclusions were produced by running the accompanying implementation and are my own.

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